COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE21S2/CPE-010	**SECTION:**	CPE21S2
DATE PERFORMED:	08/26/2026	**DATE SUBMITTED:**	08/26/2026
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Prelim Exam **[Set D]**

1. PROCEDURE OUTPUT

Supposed Algorithm of Problem Definition:

1.) Initialize an empty array-based Stack and an empty linked list-based Queue
2.) Prompt the user to input the sequence of positive integers
3.) For each integer in the input sequence, Push the integer onto the Stack.
4.) Enqueue the integer into the Queue.
5.) Record the total number of elements
6.) Set a boolean flag isSymmetric to true. Loop for how many times (or while the Stack is not empty):
7.) Pop the top element from the Stack (Dequeues the element in reverse order)
8.) Dequeue the front element from the Queue (This retrieves the elements in original order)
9.) Compare the popped value and the dequeued value (Since the values has been retrieved in the original and reversed order, it should be able to tell if it's symmetrical or not)
10.) If the values are not equal then set isSymmetric to false.
11.) If isSymmetric is true, display "The sequence is symmetric.". If isSymmetric is false, display "The sequence is not symmetric.

Pseudocode:

```
#Initialize an empty stack and queue
Class ArrayStack #Stack
Template <typename T> #Queue

For (EachInteger) {
    Output “Enter an integer”
    Input >> Stack
}

Enqueue(IntegerValue);

Create functionToRecordNumberofElements;

bool isSymmetric
    For (Number of elements in the queue) {
     Loop depending on how many values in the queue are there
}
```

```
Pop(TopElement) #In the stack
Dequeue(LastElement) #in the Queue

If  (Pop(TopElement) == Dequeue(LastElement) ) {
    isSymmetric = True;
}

If (isSymmetric = True) {
  Output “The values are symmetrical”
}

else {
Output “The values are not symmetrical”
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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